

Fashion-MNIST Classification

Using Multilayer Perceptron (MLP)

Neural Networks Course — Project README

Problem Description

This project addresses the task of **image classification** on the *Fashion-MNIST* dataset. The goal is to build a Multilayer Perceptron (MLP) that correctly identifies one of **10 clothing categories** from a 28×28 grayscale image.

Classes

Label	Class
0	T-shirt / Top
1	Trouser
2	Pullover
3	Dress
4	Coat
5	Sandal
6	Shirt
7	Sneaker
8	Bag
9	Ankle Boot

Preprocessing

- Images converted to tensors via `transforms.ToTensor()`.
- Pixel values normalized to $[-1, 1]$ using `mean = 0.5`, `std = 0.5`.
- Dataset split: **60,000** training samples / **10,000** test samples.

Dataset Link

The dataset is loaded automatically by PyTorch's `torchvision` library. No manual download is required.

<https://github.com/zalandoresearch/fashion-mnist>

Official Torchvision documentation:

<https://pytorch.org/vision/stable/generated/torchvision.datasets.FashionMNIST.html>

Experiments & Results

Three MLP configurations were trained and compared. Each model was evaluated on the **test set** (10,000 samples).

Model Architectures

Model 1 — Baseline (SGD + ReLU)

- Architecture: $784 \rightarrow 128 \rightarrow 64 \rightarrow 10$
- Activation: ReLU
- Optimizer: SGD ($lr = 0.01$)
- Epochs: 15
- Accuracy: 86.35

Model 2 — Regularized (Adam + ReLU + BatchNorm + Dropout)

- Architecture: $784 \rightarrow 128 \rightarrow 64 \rightarrow 10$
- Activation: ReLU
- Regularization: BatchNorm1d + Dropout (0.3)
- Optimizer: Adam ($lr = 0.001$), StepLR scheduler
- Epochs: 15
- Accuracy: 88.77

Model 3 — Deeper (Adam + Tanh + BatchNorm + Dropout)

- Architecture: $784 \rightarrow 512 \rightarrow 256 \rightarrow 10$
- Activation: Tanh
- Regularization: BatchNorm1d + Dropout (0.3)
- Optimizer: Adam ($lr = 0.01$), StepLR scheduler
- Epochs: 20
- Accuracy: 88.74

Comparison Table

Table 1: Performance comparison of the three MLP experiments

Model	Activation	Optimizer	Epochs	Regularization	Hidden Neurons
Model 1 (Baseline)	ReLU	SGD	15	None	128, 64
Model 2	ReLU	Adam	15	BatchNorm + Dropout	128, 64
Model 3	Tanh	Adam	20	BatchNorm + Dropout	512, 256

Note: Replace the accuracy values in the table with the exact numbers printed by the notebook after running all three models.

Instructions for Running the Project

Requirements

Install the required Python packages:

```
1 pip install torch torchvision matplotlib numpy
```

- Python 3.x
- PyTorch
- Torchvision
- Matplotlib
- NumPy

Running the Notebook

1. Clone or download the repository.
2. Open the notebook:

```
1 jupyter notebook project_nn.ipynb
```

3. Run all cells sequentially (*Kernel* → *Restart & Run All*).
4. The dataset will be downloaded automatically on first run to `./data/`.
5. Training progress and test accuracy are printed after each experiment.
6. Loss and accuracy plots are displayed inline after each model.

```
1 git clone https://github.com/Hossam293/fashion-mnist-mlp.git
2 cd fashion-mnist-mlp
```

Expected Output

Each model prints per-epoch training loss and accuracy, followed by the final test accuracy, for example:

```
1 Epoch 1: Loss = 0.8423, Acc = 70.15%
2 ...
3 Epoch 15: Loss = 0.3201, Acc = 88.42%
4 Test Accuracy: 88.35%
```

The best model

```
1 Epoch 1: Loss = 0.4613, Acc = 83.45%
2 ...poch 15: Loss = 0.3201,
3 Epoch 20: Loss = 0.2617, Acc = 90.36%
4 Test Accuracy: 88.74%
```